

Subject: Urgent: ZONE_WEST Sensor Degradation and Masked Operational Risk

Context: > Over the past week, the raw sensor logs for ZONE_WEST indicated severe, erratic pressure volatility, with readings constantly oscillating between 100 and nearly 300 PSI. On the surface, this unverified data presented an alarming picture of an unstable physical system, making accurate safety compliance and real-time monitoring practically impossible for our shift supervisors.

Insight: > After applying a rigorous data-cleaning pipeline to filter out extreme physical anomalies and extract the true 24-hour operational trend, a completely different reality emerged. The physical pressure in ZONE_WEST is highly stable, holding consistently and safely at approximately 200 PSI. The extreme volatility seen in the raw logs is an illusion caused by severe sensor degradation and high-frequency misfires, rather than an actual fluctuation in our pipeline infrastructure.

Action: > I strongly recommend deploying a maintenance team during the next available downtime to recalibrate or entirely replace the physical pressure sensors in ZONE_WEST. While the plant's pressure system is currently operating safely, relying on these highly degraded, noisy sensors creates a critical operational blind spot that could mask a genuine pressure emergency in the future.

